

Shubham

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Roll No. \_\_\_\_\_

MID-SEMESTER EXAMINATION MARCH -2024

B. TECH SEMESTER

FUNDAMENTAL OF STATISTICS AND AI

Biotech 6<sup>th</sup> Sem

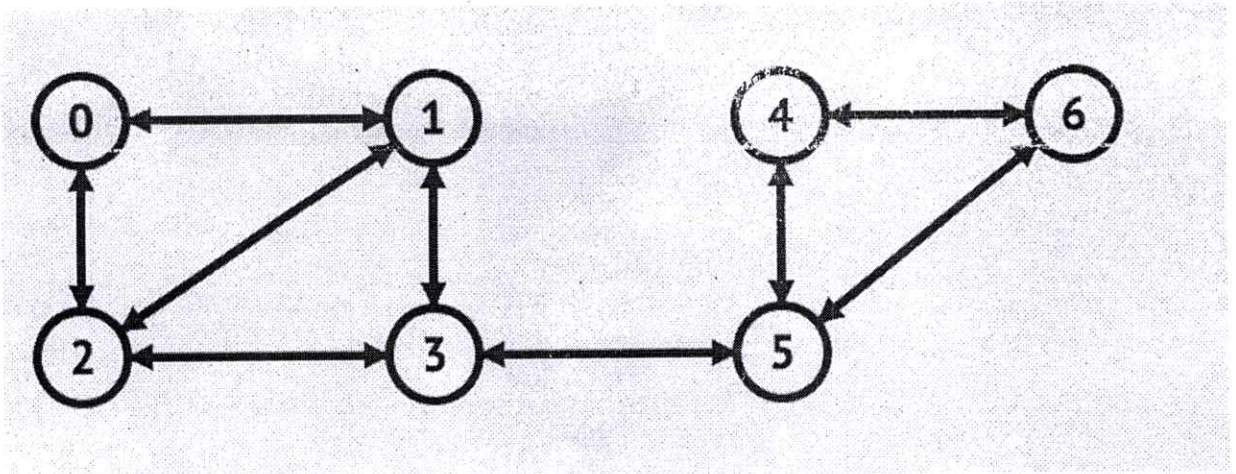
TCS-421

Time: 90 minutes

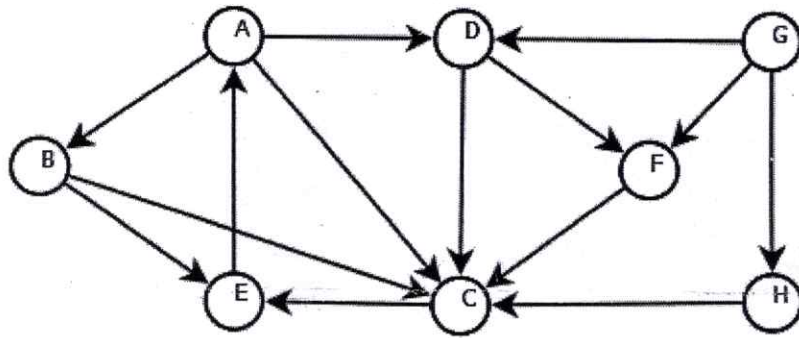
Maximum Marks: 50

Note:

- (i) Answer **all the questions** by choosing **any one of the sub-questions**.
  - (ii) Each question carries 10 marks.
1. (a). How can AI be leveraged to address real-world problems in various industries such as healthcare, finance, or education? 10 marks (CO1)
- or
- (b). Explain the working of the BFS Method and find the path to reach the goal for the following graph by applying BFS. Consider start node = 0 and goal node = 6: 10 marks (CO3)



2. (a). Explain the Turing test for artificial intelligence (AI). Describe the key components and objectives of the test. Additionally, discuss the significance of the Turing test in evaluating the capabilities of AI systems. 10 marks (CO1)
- or
- (b). Discuss the working of the DFS and apply the same to the following graph. Consider start node = A and goal node = H : 10 marks (CO3)



3. (a). Discuss the various types of Artificial Intelligence (AI) and provide examples for each type. 10 marks (CO2)

or

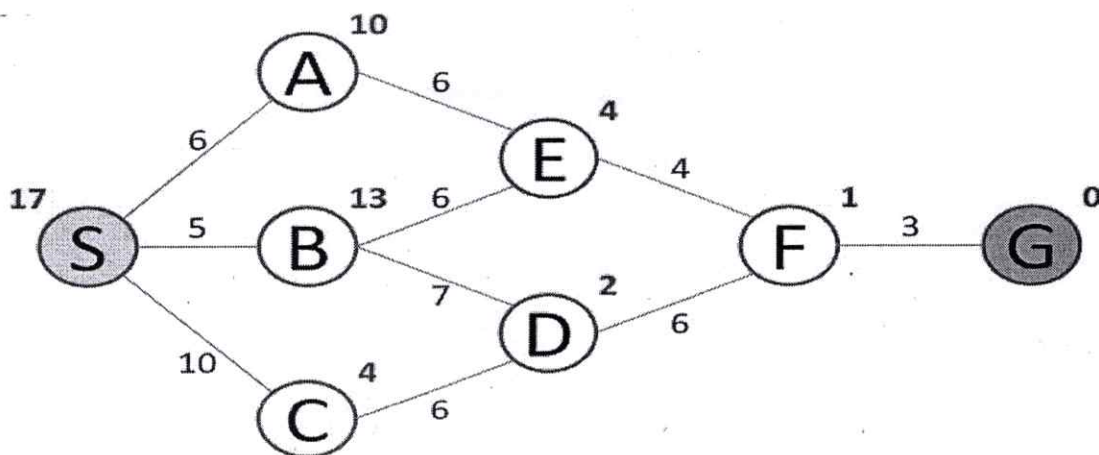
- (b). Elaborate depth limited search with suitable examples. 10 marks (CO2)

4. (a). Examine the stages and components of the Artificial Intelligence (AI) pipeline. Discuss the significance of each stage, from data collection and preprocessing to model training and deployment. 10 marks (CO2)

or

- (b). What is an agent? Explain agent types and their role in an artificially intelligent system. 10 marks (CO1)

5. (a). Perform the A\* Algorithm on the following figure. Explicitly write down the queue at each step. 10 marks (CO3)



- (b). Explain the different types of environment features. 10 marks (CO1)

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**TCS-630**

**B. TECH. (CSE/SPL) (SIXTH SEMESTER)**  
**MID SEMESTER EXAMINATION, March, 2024**

**CYBER THREAT ANALYSIS**

**Time : 1½ Hours**

**Maximum Marks : 50**

**Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.  
(ii) Each sub-question carries 10 marks.

1. (a) Describe the Dolev-Yao (DY) threat model. How is it helpful to design a secure data transmission protocol ? (CO1)

OR

- (b) Describe the Canetti and Krawczyk's adversary model (CK-adversary model). How is it helpful in the computation of a secure session key for secure data exchanges ? (CO1)

2. (a) Explain the following types of cyber threats : (CO2)  
Social engineering attacks and Distributed Denial-of-Service (DDoS) attacks. (CO2)

OR

- (b) Demonstrate the working mechanism of the Advanced Persistent Threats (APT). Use a figure in your answer. How can the APT be mitigated. (CO2)

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(2)

3. (a) Describe the following types of threat agents : (CO3)  
Organized crime groups and state-sponsored.

OR

- (b) Investigate the procedure of the following types of password-guessing attacks : (CO4)

Dictionary attack and Deployment of keylogger. (CO4)

4. (a) Demonstrate the working mechanisms of the following types of intrusion detection procedures : (CO3)

Signature-based detection and Anomaly-based detection.

OR

- (b) Explain the model for network security. Use a figure in your answer.

(CO3)

5. (a) Explain the working of a honeypot. How would you compare a pure honeypot and a high-interaction honeypot ? (CO5)

OR

- (b) How would you compare the symmetric and asymmetric key cryptographic systems ? (CO5)



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**TCS-627**

**B. TECH. (CSE) (SIXTH SEMESTER)**  
**MID SEMESTER EXAMINATION, March, 2024**

**INFORMATION RETRIEVAL AND NLP**

**Time : 1½ Hours**

**Maximum Marks : 50**

**Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.  
(ii) Each sub-question carries 10 marks.

1. (a) Explain the concept of regular expressions in the context of Natural Language Processing (NLP). Provide examples of how regular expressions can be used to extract meaningful information from text data.  
(CO2, CO3)

OR

- (b) Describe the basic steps of Natural Language Processing (NLP) including tokenization, stemming, lemmatization, POS tagging, named entity recognition, and chunking. Explain the significance of each step in the NLP pipeline.  
(CO2, CO3)
2. (a) Explain the benefits of removing stop words. Give examples in which stop words elimination may be harmful.  
(CO2, CO4)

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(2)

OR

- (b) Illustrate Parts of speech tagging and explain categories of POS tagging.  
(CO2, CO4)

3. (a) Critically evaluate the challenges and ethical considerations associated with sourcing data for NLP through web scraping. Provide recommendations for responsible data collection practices. (CO2, CO3)

OR

- (b) Discuss the process of linguistic analysis and its significance in NLP. Explain with suitable examples. (CO2, CO3)

4. (a) Define the Bag of Words model and explain its implementation in Python using the NLTK library. (CO2, CO4)

OR

- (b) Describe the linguistics properties that contribute to syntactic and semantic analysis in Natural Language Processing. (CO2, CO4)

5. (a) Compare and contrast the features, functionalities, and advantages of two popular NLP libraries : SpaCy and NLTK. Discuss scenarios where one library might be preferred over the other. (CO1, CO4)

OR

- (b) Why is Natural Language Processing (NLP) considered challenging ? Discuss the role of ambiguity in language and how does it complicate the task of NLP systems ? (CO1, CO4)



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**TCS-620**

**B. TECH. (CSE) (SIXTH SEMESTER)**

**MID SEMESTER EXAMINATION, March, 2024**

**DATA MINING FOR SMART STRUCTURES AND SYSTEMS**

**Time : 1½ Hours**

**Maximum Marks : 50**

**Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Define data mining and explain in detail about the data warehouse architecture with a neat diagram. (CO1)

OR

- (b) What is KDD ? Explain about data mining as a step in the process of knowledge discovery. (CO1)

2. (a) Why data reduction is required ? Discuss some data reduction strategies with examples. (CO2)

OR

- (b) Describe various OLAP operations using suitable examples. (CO3)

3. (a) What do you mean by data cleaning ? Describe various methods for handling "missing values" in dataset. (CO2)

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(2)

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OR

(b) Summarize the smoothing techniques followed in data cleaning process.

(CO2)

4. (a) Explain the decision tree induction algorithm with appropriate examples. Also discuss various attribute selection measures. (CO5)

OR

(b) Classify the tuple  $X = \{\text{color}='RED', \text{type}='SUV', \text{Origin}='Domestic'\}$  using Naïve Bayesian Classification. Training data is given in the following table where class label is {STOLEN} : (CO5)

| Color  | Type   | Origin   | Stolen |
|--------|--------|----------|--------|
| Red    | Sports | Domestic | Yes    |
| Red    | Sports | Domestic | No     |
| Red    | Sports | Domestic | Yes    |
| Yellow | Sports | Domestic | No     |
| Yellow | Sports | Imported | Yes    |
| Yellow | SUV    | Imported | No     |
| Yellow | SUV    | Imported | Yes    |
| Yellow | SUV    | Domestic | No     |
| Red    | SUV    | Imported | No     |
| Red    | Sports | Imported | Yes    |

(3)

5. (a) How does a Naive Bays classifier work ? Explain with an example.

(CO5)

OR

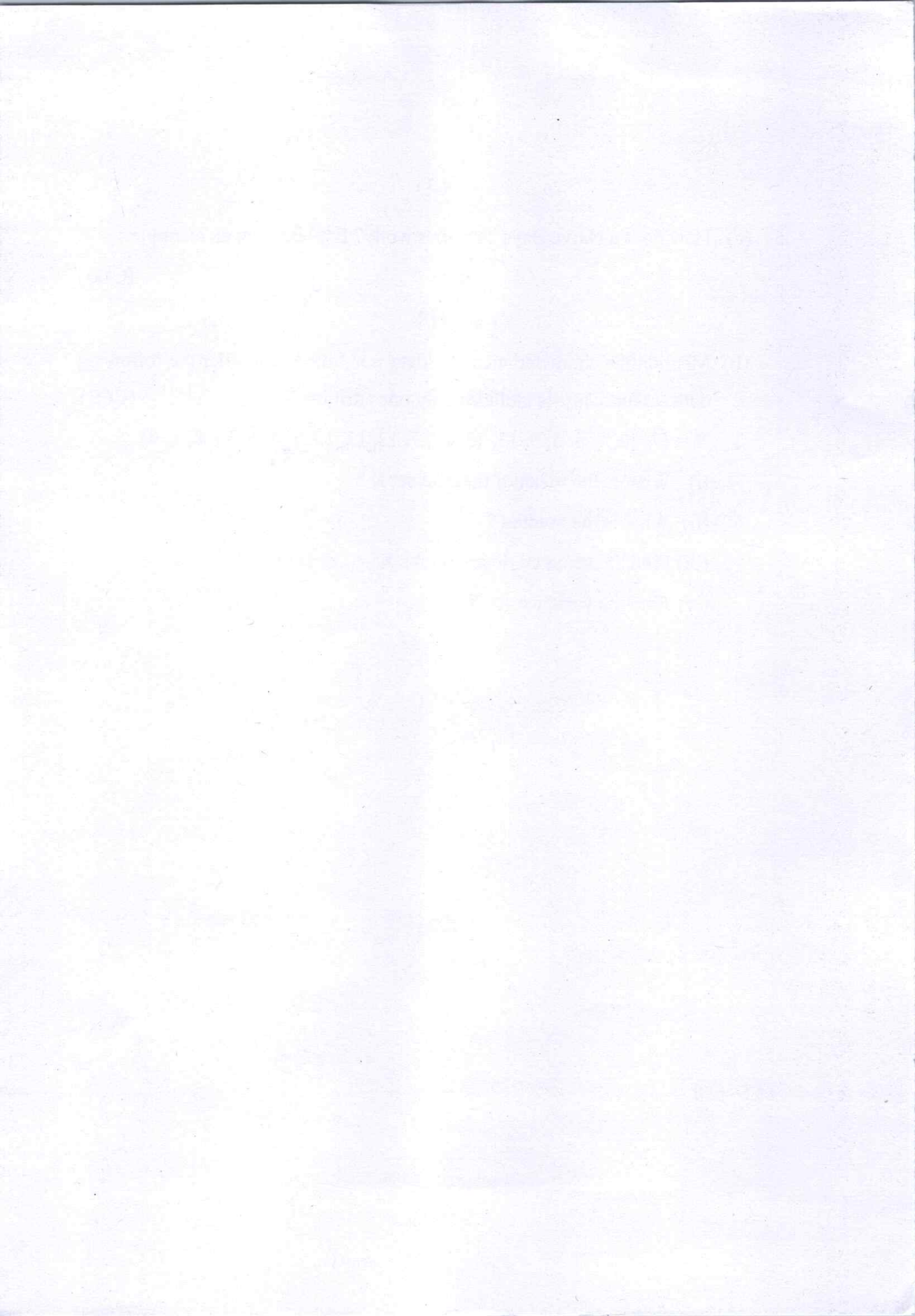
- (b) Mention the characteristics of a data warehouse. Consider the following data set for analysis includes only one attribute X :

(CO3)

$X = \{7, 12, 5, 8, 5, 9, 13, 12, 19, 7, 12, 12, 13, 3, 4, 5, 13, 8, 7, 6\}$

- (i) What is the mean of the data set X ?
- (ii) What is the median ?
- (iii) Find the standard deviation for X.
- (iv) Find the variance for X.







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**TCS-622**

**B. TECH. (CSE) (SIXTH SEMESTER)**  
**MID SEMESTER EXAMINATION, March, 2024**  
**CLOUD COMPUTING TECHNOLOGIES**

**Time : 1½ Hours**

**Maximum Marks : 50**

**Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.  
(ii) Each sub-question carries 10 marks.

1. (a) DreamCasino provides digital services to casino and sports betting services. Headquartered in Norway and founded in 2020, the company is expanding its services globally. Evaluate *four* main types of cloud computing in above mentioned organization. (CO1)

OR

- (b) Navan is a travel and expense software company that simplifies booking, viewing and managing business travel and expenses for thousands of companies around the world. Evaluate Cloud Computing Taxonomy with suitable diagram in above mentioned organization.

(CO1)

2. (a) Veriff is the preferred identity verification partner for the world's biggest and best digital companies, including pioneers in fintech,

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gaming and the mobility sectors. It provides advanced technology, deep insights and expertise from its foundation in digital-first Estonia. Evaluate the characteristics of Cloud Computing in above mentioned organization. (CO3)

OR

- (b) Prime Video is a subscription-based, on-demand over-the-top streaming service, offered as part of a Prime subscription or as a stand-alone service. Launched in 2006, it has over 200 million customers. Evaluate nine steps evolution of Cloud Computing in above mentioned organization. (CO2)

3. (a) Mediastream is a company that helps media and large companies to use streaming technologies easily. The company has been in the market for more than 15 years, with customers throughout Latin America, part of Europe and the U.S. Evaluate benefits of SaaS in above mentioned organization. (CO3)

OR

- (b) Gilead Sciences Inc. (Gilead) modernized its data infrastructure and improved its operational performance through a data mesh framework built on AWS, reducing data search times by over 50 percent. Evaluate benefits of IaaS in above mentioned organization. (CO3)
4. (a) Neeva, an ad-free search engine, uses Karpenter and Amazon EC2 Spot Instances to maintain cost optimization and reduce the time spent waiting on infrastructure management by 10—100 hours per week. Evaluate the benefits of SaaS with suitable diagram in above mentioned organization. (CO2)



(3)

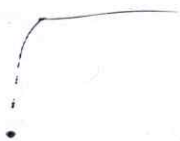
OR

- (b) Merck KGaA, Darmstadt, Germany improves its time to data insights with a connected data and analytics platform on AWS, helping data teams access secure, custom environments faster. Evaluate the importance of Open Cloud Standards with suitable diagram in above mentioned organization. (CO3)

5. (a) A division of News Corp., Dow Jones is a global provider of news and business information, delivering content to consumers and organizations around the world across multiple formats, including print, digital, mobile and live events. Evaluate how do 'standards collaborations' differ from 'open source collaborations' in above mentioned organization. (CO3)

OR

- (b) Fatshark, a Swedish video game developer, wanted to build its most complex game yet—Warhammer 40,000 : Darktide. To build on the success of the studio's Warhammer : Vermintide series, the combat-focused cooperative multiplayer game must offer ultralow latency to over 1,00,000 concurrent players. Evaluate IaaS challenges with suitable diagram in above mentioned organization. (CO1)





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**TCS-619**

**B. TECH. (CSE & SPL) (SIXTH SEMESTER)**  
**MID SEMESTER EXAMINATION, March, 2024**

**NETWORK AND SYSTEM SECURITY**

**Time : 1½ Hours**

**Maximum Marks : 50**

**Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Describe the following attacks defined under the OSI security architecture : (CO1)

Interruption Interception and Modification

OR

- (b) Demonstrate that a secure communication system can be designed through the guidelines of the model for network security. (CO1)

2. (a) Describe the following security services : (CO2)

Data integrity, Access control and Non-repudiation

OR

- (b) Draw and explain the single round of the Data Encryption Standard (DES) algorithm. What is the role of placing permuted choice 2 (PC-2) in this architecture ? (CO2)

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(2)

3. (a) Describe the following general security requirements : (CO4)

Data freshness, Authentication and Data availability

OR

- (b) Investigate the encryption and decryption procedures of the triple DES algorithm with three keys. What are the applications of this algorithm ?

(CO4)

4. (a) Use the decryption procedure of the affine cipher scheme and decipher the ciphertext "HPCCX AQ" if the encryption equation is  $E(x) = (5x + 8) \bmod 26$ . (CO3)

OR

- (b) Investigate that the cipher block chaining (CBC) mode of operation of block cipher is vulnerable to bit-flipping attacks. (CO3)

5. (a) State and prove the Vernam cipher is a perfectly secure cryptosystem.

(CO5)

OR

- (b) Demonstrate the encryption and decryption procedures of the RSA algorithm through an example. (CO5)



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**TCS-671**

**B. TECH. (CSE) (SIXTH SEMESTER)**

**MID SEMESTER EXAMINATION, March, 2024**

**BIG DATA STORAGE AND PROCESSING**

**Time : 1½ Hours**

**Maximum Marks : 50**

**Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Discuss Big Data, its types and five dimensions of Big Data with examples. (CO1)

OR

- (b) Explain the importance of data organization and data architecture in the context of Big Data. (CO1)

2. (a) Elaborate real-world examples where Big Data analytics has led to significant improvements or innovations. (CO5)

OR

- (b) Describe the challenges associated with the storage and management of large volumes of data. (CO5)

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(2)

3. (a) Describe the various techniques and tools used for analyzing Big Data.

(CO2)

OR

- (b) Enumerate the key features of the MapReduce programming model in Hadoop.

(CO2)

4. (a) Explain the key design principles of HDFS and how they contribute to its scalability and fault tolerance.

(CO3)

OR

- (b) Discuss the impact of a Name Node failure on the overall availability and reliability of an HDFS cluster.

(CO3)

5. (a) Describe the functions of Name Node, Data Nodes, Job Tracker and Task Tracker in HDFS and their responsibilities in the distributed storage architecture.

(CO6)

OR

- (b) Discuss how HDFS handles node failures and ensures data integrity in the presence of hardware or network issues.

(CO6)



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**TCS-651**

**B. TECH. (CSE) (SIXTH SEMESTER)**  
**MID SEMESTER EXAMINATION, March, 2024**

**DEVOPS ON CLOUD**

**Time : 1½ Hours**

**Maximum Marks : 50**

**Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Explain the concept of a DevOps. Discuss components and practices that are typically included in such as environment. (CO2, CO3)

OR

- (b) Describe the key advantages and benefits adopting cloud computing for businesses or organizations ? How does the implementation of DevOps principles impact the efficiency and scalability of cloud-based infrastructure ? (CO2, CO3)

2. (a) How does the adoption of DevOps principles lead to increased collaboration and communication among development, operations, and quality assurance teams ? (CO1, CO2)

OR

- (b) Discuss the key principles of the Agile Framework and how they contribute to successful software development projects. (CO1, CO2)

**P. T. O.**



(2)

3. (a) Discuss the differences between DevOps and Agile methodologies. How do they complement each other in Software development processes ? (CO1, CO3)

OR

- (b) Define Software Engineering. Explain its importance in the development of complex Software systems. Compare and contrast Waterfall and Agile methodologies in Software development. Highlight the advantages and disadvantages of each. (CO1, CO3)
4. (a) Describe the basic workflow in Git version control. Discuss concepts such as repositories, commits, branches and merges. (CO2)

OR

- (b) Discuss version control, and why is it essential in Software development. Explain the benefits of using version control systems like Git. (CO2)
5. (a) Explain the concept of Continuous Integration (CI) and how Jenkins facilitates Continuous Integration (CI) practices, additionally discuss the key features of Jenkins that make it a popular choice for automation and CI/CD. (CO2)

OR

- (b) Describe the DevOps Ecosystem. List and briefly explain tools and technologies are commonly used in DevOps practices viz. deployment, automation, CI/CD, monitoring, alerting, version control and scripting. (CO2)



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**TCS-629**

**B. TECH. (CSE) (SIXTH SEMESTER)**  
**MID SEMESTER EXAMINATION, March, 2024**

**ARTIFICIAL INTELLIGENCE**

**Time : 1½ Hours**

**Maximum Marks : 50**

**Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain intelligence and artificial intelligence. How do you distinguish between the two ? How are search strategies used for problem solving in AI ? (CO1)

OR

- (b) What are intelligent Agents ? Explain the properties of the Environment in designing Intelligent Agents. (CO2, CO3)

2. (a) Discuss the depth-first and breadth-first searching methods. Suggest some applications to which each is best suited ? (CO1)

OR

- (b) What is the idea of heuristics search in AI ? Write a note on AI programming languages. (CO2)

**P. T. O.**



(2)

3. (a) Explain AI Agents design philosophy using neat and clean diagrams.

(CO3)

OR

- (b) What is the Turing Test in Artificial Intelligence ? Explain. (CO2)

4. (a) Convert the following in First order predicate logic : (CO3)

(i) All birds fly.

(ii) Every man respects his parents.

OR

- (b) Discuss nearest neighbor classification with a suitable example.

(CO2)

5. (a) What is the fundamental structure of an expert system ? Provide instances that are appropriate to illustrate its usefulness in various fields.

(CO2/CO3)

OR

- (b) Discuss Hill Climbing search algorithm.

(CO1)



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**TCS-691**

**B. TECH. (CSE) (SIXTH SEMESTER)**  
**MID SEMESTER EXAMINATION, March, 2024**  
**IMAGE PROCESSING AND COMPUTER VISION**

**Time : 1½ Hours**

**Maximum Marks : 50**

**Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) How digital image processing techniques can address issues like occlusion, varying lighting conditions, and rapid changes in the real time computer vision applications ? Explain with an example. (CO1)

OR

- (b) Discuss the concept of power-law transformation. Also provide a scenario where power-law transformation could be beneficial. (CO1)

2. (a) Why do we need Bit-plane slicing in image processing ? Consider a 5×5 image matrix and show each and every step. (CO1)

OR

- (b) How are sampling and quantization essential processes in the digitization of continuous image data ? Justify your answer. (CO1)

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(2)

3. (a) Consider a  $5 \times 5$  image matrix and perform edge detection method using a  $3 \times 3$  SOBEL operator. Show each and every step of edge detection processing. (CO2)

OR

- (b) What are the benefits of Otsu thresholding ? Consider a  $5 \times 5$  below image matrix and show the processing of Otsu thresholding. (CO2)
4. (a) How sharpening filter works on the blurry images ? Explain each and every step by apply  $3 \times 3$  sharpening filter on  $5 \times 5$  blurry image matrix. (CO2)

OR

- (b) Consider a  $5 \times 5$  below 256 grey level image matrix and compute the equalized histogram : (CO2)

|    |     |    |    |     |
|----|-----|----|----|-----|
| 25 | 100 | 75 | 49 | 130 |
| 50 | 80  | 0  | 70 | 100 |
| 5  | 10  | 20 | 30 | 0   |
| 60 | 50  | 12 | 24 | 32  |
| 37 | 53  | 55 | 21 | 90  |

5. (a) How average operations are performed in Digital image processing ? Perform each and every step by apply average operations on  $5 \times 5$  blurry image matrix. (CO3)

OR

- (b) What is the significant of negative image ? Explain each and every step by considering  $5 \times 5$  blurry image matrix to obtain negative image. (CO3)



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**TCS-642**

**B. TECH. (CSE) (SIXTH SEMESTER)**  
**MID SEMESTER EXAMINATION, March, 2024**

**BIG DATA ANALYTICS**

**Time : 1½ Hours**

**Maximum Marks : 50**

**Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What is Big data analytics ? Explain five V's of Big data. Briefly discuss applications of Big data. (CO1)

OR

- (b) Differentiate between Structure and Unstructured data. What are the different sources of Structure and Unstructured data ? (CO1)

2. (a) What is data sampling in terms of Big data analytics ? Discuss at least *five* types of sampling techniques. (CO1)

OR

- (b) What are the top challenges facing Big data ? Why is Big data important ? Why is an e-mail placed in the "unstructured data category" ? (CO1)

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(2)

3. (a) Define in-memory analytics and explain how it differs from traditional disk-based analytics. What are the key advantages of using in-memory analytics in terms of speed and performance ? (CO4)

OR

- (b) Discuss the concept of streams in the context of data processing. Discuss the Stream Data Model and its architecture. (CO4)

4. (a) Discuss the concept of Regression Analysis, focusing on the demand curve and making predictions. How can regression analysis be applied to practical business scenarios ? (CO1)

OR

- (b) What is Hadoop ? Explain the core components of the Hadoop ecosystem. What are the *three* modes in which a Hadoop application can be executed ? (CO1)

5. (a) Provide an overview of the *three* main types of analytics : descriptive, predictive and prescriptive. How do these types of analytics differ in terms of their objectives and methodologies ? (CO3)

OR

- (b) How does media planning contribute to marketing strategies ? Provide examples of how analytics can be applied in media planning to optimize advertising campaigns. (CO3)



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**TCS-672**

**B. TECH. (CSE) (SIXTH SEMESTER)**  
**MID SEMESTER EXAMINATION, March, 2024**  
**DEEP LEARNING WITH KERAS AND TENSORFLOW**

**Time : 1½ Hours**

**Maximum Marks : 50**

**Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.  
(ii) Each sub-question carries 10 marks.

1. (a) What do you understand by Artificial Intelligence ? Explain with the help of the illustrative example. (CO1)

OR

- (b) Discuss the difference between unsupervised learning and supervised learning. Provide examples to illustrate your explanation. (CO1)

2. (a) Explain what is vanishing and exploding gradient problem in training deep neural networks. (CO2)

OR

- (b) Describe the advantages of using Keras as a user-friendly API on top of TensorFlow. (CO2)

**P. T. O.**



(2)

3. (a) Discuss the application of deep learning in computer vision tasks. What do you think are the major challenge in training deep neural network for vision task. (CO1, CO2)

OR

- (b) Explain the concept of the backpropagation algorithm in neural network learning. Describe the key steps involved in backpropagation, including forward propagation, error calculation, and weight updates. (CO1/CO2)
4. (a) Designing a health care system using deep neural networks. Discuss in detail all the steps involved in designing such system. (CO2, CO3)

OR

- (b) Explain the concept of the gradient descent optimization algorithm. What are the disadvantages of gradient descent algorithm ? (CO2, CO3)
5. (a) What is the significance of the learning rate in training machine learning models. Discuss how the learning rate affects the convergence of optimization algorithms such as gradient descent. (CO2, CO3)

OR

- (b) Explain what is the problem of overfitting in training deep neural networks. Name *one* type of regularization to circumvent the effect of overfitting. (CO2, CO3)



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**TCS-611**

**B. TECH. (CSE) (SIXTH SEMESTER)**

**MID SEMESTER EXAMINATION, March, 2024**

**SOFTWARE ENGINEERING**

**Time : 1½ Hours**

**Maximum Marks : 50**

- Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.  
(ii) Each sub-question carries 10 marks.

1. (a) Explain the Waterfall model and the Agile approach, highlighting their key differences, advantages, and disadvantages. (CO1)

OR

- (b) Explain the concept of software testing and its importance in the software development lifecycle. Discuss different levels of software testing. (CO1)

2. (a) Define the concept of the software crisis and discuss its underlying causes. Provide examples of historical incidents or projects that illustrate the challenges associated with the software crisis. (CO2)

**P. T. O.**



OR

- (b) Discuss Requirement Engineering (RE) process. Elaborate various challenges associated with requirement elicitation. (CO2)
3. (a) Explain the significance of a Software Requirements Specification (SRS) document in the software development process. Outline the key components typically included in an SRS document. (CO3)

OR

- (b) Bank X is planning to develop a mobile banking application to provide its customers with convenient access to banking services on their smartphones. Your task is to identify functional and non-functional requirements for the mobile banking application. (CO3)
- (i) Identify five functional requirements that should be included in the mobile banking application. Justify your choices with examples of typical banking transactions and functionalities that users would expect from the app.
- (ii) List three non-functional requirements that are crucial for the security and reliability of the mobile banking application. Explain how these non-functional requirements help protect sensitive customer data and ensure uninterrupted service availability.
4. (a) Explain the key characteristics of software and their significance in the development and maintenance of software systems. Provide examples to illustrate how each characteristic influences the behavior, quality, and evolution of software products. (CO3)



(3)

OR

(b) Design a DFD for the following : (CO3)

- (i) Borrowing a book from library
- (ii) Students registration for online course in a university

5. (a) Elaborate phases and workflows in Rational Unified Process (RUP) model. Also discuss the purpose of iterations in software development and how they contribute to the success of RUP projects. (CO2)

OR

(b) Discuss the concept of a decision table and its role in representing conditional logic in software systems. Provide an example of a decision table for a simple business scenario showing how different combinations of conditions and actions are represented in the table.

(CO2)



(1)

OK

(C2)

(b) Design a DFD for the following

(c) Borrowing a book from library

(d) Student registration for online course in university

(e) Reserve places and workflow in National United Bank (RUB)

model. Also discuss the purpose of questions in your development

and how they contribute to the success of RUP projects

(C2)

OK

(f) Discuss the concept of a business process and its role in representing

business processes in a business process model. Provide an example of a business

process model and discuss its components, showing how different

components of a business process model are related to the tasks

(C2)

1000

1000-01



Roll No. ....

**TCS-601**

**B. TECH. (ALL BRANCHES)  
(SIXTH SEMESTER)**

**MID SEMESTER EXAMINATION, March, 2024**

**COMPILER DESIGN**

**Time : 1½ Hours**

**Maximum Marks : 50**

- Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.  
(ii) Each question carries 10 marks.

1. (a) Write the operator precedence parsing algorithm. Obtain the precedence relation table for the following grammar : (CO1)

$E \rightarrow \dot{E} \text{ or } E | E | E \text{ and } E | \text{not } E | (E) | \text{true} | \text{false}$

**OR**

- (b) List the significant responsibilities of a lexical analyzer. Count the number of tokens in the following code segment : (CO1)

```
int fun(in n) {  
    in x=0, c;  
    while(c!=0) {  
        if(c & 01) x++;  
        else c>>=1; }  
    return c; }
```

**P. T. O.**



2. (a) What are the different phases of a compiler ? Pass the string 'int a=b + c\*10.0; from each phase of the compiler and show the output. (CO1)

OR

- (b) Check for the LR (0) by making a table of the following grammar :

(CO1)

$S \rightarrow ASB \mid AB$

$A \rightarrow a$

$B \rightarrow b$

3. (a) Prove that the given grammar G is CLR but not SLR (1). (CO2)

$S \rightarrow L = R \mid R$

$L \rightarrow * R \mid id$

$R \rightarrow L$

OR

- (b) Explain the task of the semantic analyzer. Also, compare the working of static and dynamic checkers. (CO2)

4. (a) Why do parsers compute First and Follow functions ? Compute the First and Follow for the given grammar : (CO2)

$S \rightarrow AaB \mid BbA$

$A \rightarrow \epsilon \mid Aa \mid Bb$

$B \rightarrow \epsilon \mid Aa \mid Bb$

Compute FIRST and FOLLOW sets.



(3)

OR

- (b) Analyze the running cost of a single input buffering scheme. State the advantages of the double buffering scheme over the single buffering scheme. (CO2)
5. (a) Perform recursive decent parsing for the following grammar : (CO2)
- Expr  $\rightarrow$  Expr + Term | Term
- Term  $\rightarrow$  Term \* Factor | Factor
- Factor  $\rightarrow$  (Expr) | num

OR

- (b) Draw the block diagram of the predictive parser (PP). Using the given grammar, construct a predictive parser table and parse the input string id%id↑id\$ using the predictive parser table. (CO2)

$E \rightarrow TE'$

$E' \rightarrow \%TE' / \epsilon$

$T \rightarrow FT'$

$T' \rightarrow \uparrow FT' / \epsilon$

$F \rightarrow id$



(1) Analyze the running cost of a light bulb. Estimate the  
advantages of the double bulb and over the single bulb  
(100)

(2) Estimate the running cost of a light bulb. Estimate the  
advantages of the double bulb and over the single bulb  
(100)

(3) Estimate the running cost of a light bulb. Estimate the  
advantages of the double bulb and over the single bulb  
(100)

(4) Estimate the running cost of a light bulb. Estimate the  
advantages of the double bulb and over the single bulb  
(100)

(5) Estimate the running cost of a light bulb. Estimate the  
advantages of the double bulb and over the single bulb  
(100)

(6) Estimate the running cost of a light bulb. Estimate the  
advantages of the double bulb and over the single bulb  
(100)

(7) Estimate the running cost of a light bulb. Estimate the  
advantages of the double bulb and over the single bulb  
(100)

(8) Estimate the running cost of a light bulb. Estimate the  
advantages of the double bulb and over the single bulb  
(100)

(9) Estimate the running cost of a light bulb. Estimate the  
advantages of the double bulb and over the single bulb  
(100)



Roll No. ....

**TCS-604**

**B. TECH. (CSE) (SIXTH SEMESTER)**  
**MID SEMESTER EXAMINATION, March, 2024**

**COMPUTER NETWORKS-I**

**Time : 1½ Hours**

**Maximum Marks : 50**

**Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Discuss the concept of encapsulation in the OSI model. Can you think of real-world analogies that effectively convey the idea of encapsulation in networking ? (CO1, CO3)

**OR**

- (b) Consider a network with four nodes (A, B, C, D) and each node wants to establish a dedicated circuit with every other node for voice communication. The network uses circuit switching and each circuit requires 64 Kbps. Assume that the setup time for each circuit is 10 seconds : (CO1, CO3)

- (i) Calculate the total number of circuits needed for full connectivity.  
(ii) Determine the total setup time required for all circuits to be established.

**P. T. O.**



2. (a) Consider a network where the transmission rate is 1 Gbps (Gigabit per second) and the average packet size is 1000 bits. The propagation delay is 10 milliseconds and the total transmission time for a packet from source to destination is 20 milliseconds. Additionally, there are four hops between the source and destination.

Calculate the following : (CO1, CO3)

- (i) Transmission delay
- (ii) Propagation delay
- (iii) Total delay
- (iv) Throughput

OR

- (b) Consider an FITTP client that wants to retrieve a Web document at a given URL. The IP address of the HTTP server is initially unknown. What transport and application-layer protocols besides HTTP are needed in this scenario ? What messages client will send and to whom to get IP address of the server ? What messages are exchanged between client and server to finally get the Web document ? Explain briefly with suitable example. (CO1, CO3)

3. (a) Suppose you have a file of 8000 bits that needs to be transmitted from a source to a destination over a packet-switched network. The network has a transmission rate of 2 Mbps (Megabits per second) and each packet can carry 1000 bits of data. The propagation delay is negligible,



and there are two intermediate routers between the source and destination. Calculate total transmission time for all the packets to reach the destination.

(CO1, CO2)

OR

- (b) How do SMTP, POP and MIME collectively contribute to the efficient delivery, retrieval and visually rich presentation of multimedia content in e-mails ? Discuss the specific responsibilities of each protocol in handling diverse media types, ensuring successful transmission and enhancing the overall user experience. (CO1, CO2)
4. (a) Suppose you needed to use HTTP to download a webpage with three embedded images. Draw diagrams, similar to those from class, depicting the main interactions between the client and server when using non-persistent HTTP, persistent HTTP without pipelining and persistent HTTP with pipelining. (CO2, CO3)

OR

- (b) Consider distributing a file of  $F = 25$  Gbits to  $N$  peers. The server has an upload rate of  $u_s = 30$  Mbps and each peer has a download rate of  $d_i = 2$  Mbps and an upload rate of  $u$ .

For  $N = 10, 100$  and  $1000$  and  $u = 300$  Kbps,  $700$  Kbps and  $2$  Mbps, prepare a chart giving the minimum distribution time for each of the combination of  $N$  and  $u$  for both client-server distribution and P2P distribution.

(CO2, CO3)



5. (a) Describe the criteria used by a peer to select which other peers to choke. Define "optimistic unchoking" and its role in BitTorrent. What purpose does optimistic unchoking serve and how does it complement the choking mechanism ? (CO2, CO1)

OR

- (b) What are the main tasks and resource records of DNS ? Discuss all the tuples used in it. (CO2, CO1)



Roll No. ....

**TCS-693**

**B. TECH. (CSE) (SIXTH SEMESTER)**  
**MID SEMESTER EXAMINATION, March, 2024**  
**FULL STACK WEB DEVELOPMENT**

**Time : 1½ Hours**

**Maximum Marks : 50**

**Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) What is a meta tag ? How meta tag works and why is meta tag important ? Benefits of using meta tag, types of meta tag. (CO1)

OR

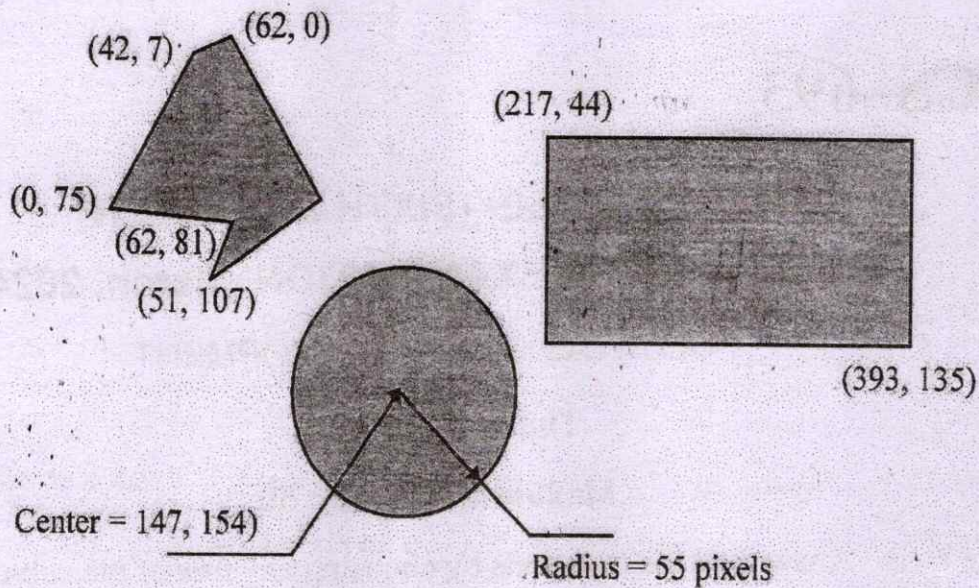
- (b) Make a distinction between the post and get methods. Additionally distinguish between buttons and input types submit. Codify the provided frameset. (CO1)

|        |        |
|--------|--------|
| Page 1 | Page 3 |
|        | Page 4 |
|        | Page 5 |
| Page 2 |        |

**P. T. O.**



2. (a) What is a list ? Explain the type of list. What is an image map ? Write the HTML code for the given map. (CO1)



OR

- (b) With appropriate examples, explain the differences between a selection and a datalist. Describe the form components in HTML 5. For the provided Table, write the HTML code. (CO1)

**TIME TABLE**

| Day/Period | I<br>9:30-<br>10:20 | II<br>10:20-<br>11:10 | III<br>11:10-<br>12:00 | 12:00-<br>12:40       | IV<br>12:40-<br>1:30 | V<br>1:30-<br>2:20 | VI<br>2:20-<br>3:10 | VII<br>3:10-<br>4:00 |
|------------|---------------------|-----------------------|------------------------|-----------------------|----------------------|--------------------|---------------------|----------------------|
| Monday     | Eng                 | Mat                   | Che                    | L<br>U<br>N<br>C<br>H | LAB                  |                    |                     | Phy                  |
| Tuesday    | LAB                 |                       |                        |                       | Eng                  | Che                | Mat                 | SPORTS               |
| Wednesday  | Mat                 | Phy                   | Eng                    |                       | Che                  | LIBRARY            |                     |                      |
| Thursday   | Phy                 | Eng                   | Che                    |                       | LAB                  |                    |                     | Mat                  |
| Friday     | LAB                 |                       |                        |                       | Mat                  | Che                | Eng                 | Phy                  |
| Saturday   | Eng                 | Che                   | Mat                    |                       | SEMINAR              |                    |                     | SPORTS               |



3. (a) Give a relevant example to illustrate the attribute of HTMLS Form elements and create the HTML code for the specified Form. (CO2)

Full Name

Date of Birth

Weekly Work Capacity

Profile Picture  No fil...hosen

Gender ☐ Male  
☐ Female  
☐ Other  
☐ I'd rather not specify

Food Choices ☐ Vegan  
☐ Vegetarian  
☐ Non Vegetarian  
☐ I'd rather not specify

Primary Language

Your Email

Your Phone Number

Your Website

Your Password

Confirm Password

☐ Terms and Conditions

OR

- (b) What does the CSS box model mean ? Which attributes of CSS are included in it ? What benefits does the use of CSS offer ? What are the various pseudo elements and pseudo classes in CSS ? (CO2)



4. (a) Using a JavaScript API method, write the code in JavaScript to display all three types of screen coordinates. (CO2, CO3)

OR

- (b) Which types of selectors are available in CSS ? What is the number of ways we can add CSS to our HTML file ? Which type of CSS holds the highest priority ? (CO2, CO3)
5. (a) Which data types are available in JavaScript ? In JavaScript, how do you construct an array ? Create a JavaScript that counts the days since the year's start. (CO3)

OR

- (b) How may JavaScript code be added to an HTML file ? Create a JavaScript function that accepts as an argument a string that contains both capital and lowercase characters. It translates letters from upper case to lower case and vice versa. (CO3)